Akash Dubey

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Education

Rutgers University Honors College, School of Arts and Sciences New Brunswick, NJ
Bachelor of Science in Mathematics and Computer Science *Anticipated May 2027*

- **GPA**: 3.95/4.0 | SAT: 1580
- **Relevant Coursework**: Math Theory of Probability, Stochastic Processes, Tensor Networks, Honors Calculus III & IV, Algorithms, Data Structures, Systems Programming, Computer Architecture.
- **Academic Programs & Research**: Directed Research Reading Program on Symmetric Functions (representation theory for combinatorial optimization); Physics Learning Theory Seminar (statistical mechanics in high-dimensional learning), Dean's List.
- **Activities**: Quantitative Finance Club, Road to Silicon Valley Program (Cohort 6), ESL Instructor, Pickleball.

Skills & Awards

Languages: C++, CUDA, Python, R, SQL, Java, TypeScript, Bash, Git
Quantitative: Probability & Stochastic Processes, Statistics, Econometrics, Optimization, Linear Algebra, Numerical Methods, High-Performance Computing (HPC)
Technologies: Python, R, SQL, C++, Statistics, Econometrics, Quantitative Modeling, XGBoost, Pandas, NumPy, PyTorch, AWS, PostgreSQL
Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.

Quantitative & Computational Research

Rutgers Economics Labs New Brunswick, NJ
President (formerly Research Director, Team Lead) *Oct 2024 – Present*

- Leading quantitative research projects for partners including NJDOL, NJDEP, NJBPU, Federal HUD, U.S. Census Bureau; applied econometric models in Python/R to analyze large datasets and project economic trends.
- Previously led a team of five researchers to evaluate the efficacy of the Urban Enterprise Zone program for the NJDCA, utilizing econometric methods in R and Python to analyze complex census datasets.
- Architected an automated economic research pipeline RAIL (Rutgers Agentic Intelligence Labs) to collect, organize, analyze, and synthesize economic data to derive insights using data ontologies and coding agent workflows.

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS New Brunswick, NJ
Research Assistant (Advisor: Prof. Jingjin Yu) *Jan 2026 – Present*

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional ASAO robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy (GaP) for dual-arm task-and-motion planning (TAMP), learning from successful and failed planner traces to rank symbolic action skeletons and predict geometric feasibility.

Kwan Lab, Rutgers Cancer Institute of New Jersey New Brunswick, NJ
Computational Research Assistant (Advisor: Prof. Kelvin Kwan) *Jan 2026 – Present*

- Built a 3D chromatin polymer simulator using overdamped Langevin dynamics to generate conformations from Hi-C contact matrices; implemented multi-scale detection via insulation score and directionality index methods and validated the pipeline with 70+ automated tests.
- Developed an in silico deletion screen spanning 12 candidate sites, 3 deletion sizes, and 2 cell types (72 perturbation conditions); resolved legacy TensorFlow/Keras dependencies on HPC for reproducible sequence-based inference.

Engineering Experience

Lykke San Francisco, CA
Founding Software Engineer *Sep 2025 – Present*

- Engineered adaptive data pipelines across MongoDB, Zilliz/Milvus, S3, and SSE streaming, including token-budgeting for large corpora and fallback paths for extracted vs. full-document reasoning.
- Built a distributed generation engine running tenant-isolated background jobs via SQS, MongoDB leases, Redis rate limiting, and AWS ECS, with embedded content for low-latency retrieval.

New York Life Insurance New York, NY
Software Engineer Intern, TDAV Team *May – Aug 2026*

- Built an AI text and voice agent for processing insurance claims across 12,000 Customer Service Representatives on AWS infrastructure with access to NYL services, knowledge, and forms.
- Constructed a knowledge base from 13,000 documents compiled across SharePoint, Confluence, Jira, and the Intranet, consolidating business logic into one queryable system.

Scarlet Sync New Brunswick, NJ
Founder *Jan 2025 – Present*

- Built a Rutgers academic planning platform with thousands of users; architected a Python data ingestion pipeline to reverse-engineer legacy data, indexing 2,500+ classes and 500+ degree programs for real-time querying.

Samaritan Scout Cranford, NJ
Full Stack Engineer Intern *May 2023 – Aug 2025*

- Built an agentic scraping system to extract structured data from 100,000+ websites and a full-stack search engine (React, TypeScript, PostgreSQL) indexing 35,000+ records with semantic search and accuracy filters.

Selected Technical Projects

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC, CUDA*

- Fine-tuned small language models on the GSM8k math reasoning dataset to study the effects of model size and architecture on mathematical reasoning.
- Deployed models on the Rutgers High Performance Computing (Amarel) cluster, utilizing quantization and structural pruning to maximize inference throughput on constrained compute.

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Retrieval*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.
- Implemented MCP/CLI and agent infrastructure for auditable workflows, including local runner dispatch, dry-run work orders, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Vision* (private code)

- Built a native desktop agent integrating vision models with macOS Accessibility APIs, AppleScript, and JXA, using the Model Context Protocol (MCP) to orchestrate specialized sub-agents for multi-app workflows.